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Real-World M3-BREATHE: Toward Multimodal Mobile Monitoring of Behaviour, Respiration, and Exposures for Treatment and Health Evaluation

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Abstract

The integration of mobile health (mHealth) technologies for continuous, real-time monitoring of physiological and behavioural signals, in concert with detailed measurement of environmental exposures, holds transformative potential for advancing personalized health risk assessment and enabling more effective treatment and intervention. In this protocol, I introduce a multimodal mobile monitoring system specifically designed to capture and integrate data on individual behaviour (daily activities and travel), respiratory function, physiological parameters and environmental exposures. The system combines mobile health devices and portable environmental sensors with GPS-enabled smartphone-based applications to generate synchronized, high-resolution data streams. These data are processed through a smart mobile phone application by incorporating physiological parameter calculation (e.g., physical activity intensity, respiratory rate) and external urban environmental data (e.g., urban green spaces, street view maps), after which a machine learning-based alert system was deployed to provide actionable insights into respiratory health, daily activities and mobility patterns, and exposure to risk factors. The protocol provides comprehensive and detailed guidance on hardware setup, software configuration, data acquisition procedures, and analytical workflows necessary for deploying the Multimodal Mobile Monitoring of Behaviour, Respiration, and Exposures (air pollution, noise, light, heat, greenness, etc.) for Treatment and Health Evaluation (M3-BREATHE) in both controlled clinical experiments and real-world contexts. This forward-looking M3-BREATHE platform establishes a scalable, non-invasive, and cost-efficient approach to longitudinal environment-mobility-health (E-M-H) monitoring, linking individual-level precision health in everyday life with long-term health intelligence at the population scale.

Introduction

Moving beyond traditional spatial epidemiology that has primarily emphasised population-level and area-based associations between environment and health, the focus is now shifting toward examining individual-level and mobility-based relationships and processes[1,2,3,4]. This shift is driven by several interrelated fundamental methodological problems, alongside emerging opportunities for more precise and context-sensitive approaches.

First, the modifiable areal unit problem (MAUP), which refers to the problem that, in the analysis of area- or zone-based data, the results may be influenced by the definition of the areal units for which data are reported[5,6]. It applies to two separate, but interrelated, problems with spatial data analysis. The first is the “scale problem”, where

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the same set of areal data is aggregated into several sets of larger areal units, with each combination leading to different data values and inferences. The second aspect of the MAUP is the "zoning problem", where a given set of areal units is recombined into zones that are of the same size but located differently, again resulting in variation in data values and, consequently, different conclusions[7,8]. In order to deal with the questioning on the study's contribution to advancing new knowledge utilizing area- or zone-based data, two methods are unremittingly discussed to minimize the impact of the MAUP. One is the maximization of within-area homogeneity. The selection of the proper areal unit of study will minimize measurement error of environmental exposure and non-differential exposure misclassification when estimating the contextual effect on health[9,10,11]. However, the heterogeneity of air pollution exposure within spatial units may still result in errors. Thus, another method is moving toward frame-independent forms of spatial data representation and analysis[12,13]. Moving beyond the conventional area-based understanding of contexts, individual-based measures account for the attribute homogeneity and spatial contiguity of contextual factors and can offer a robust analytical measure that is scale-independent or scale-invariant when examining the effects of environmental exposures on health[14,15,16].

Second, the Uncertain Geographic Context Problem (UGCoP), referring to the problem that findings about the effects of area-based contextual variables on individual behaviours or outcomes may be affected by how contextual units (e.g., neighbourhoods) are geographically delineated and the extent to which these areal units deviate from the true geographic context[17,18]. A typical example is the evaluation of individuals' exposure to air pollution may be erroneous by using residential locations for people who spend much time out of their homes for work, leisure, etc., which is far from the "true causally relevant" geographic context[18,19]. To mitigate the contextual uncertainties that influence the accurate assessment of people's environmental exposures and their health effects, it is necessary to adopt mobility-based measurement that moves beyond notions of contextual influences that rely on using specific fixed locations (e.g., residential location, workplace)[16,20]. As the neglect of spatiotemporal variability in air pollution and human mobility could lead to substantial errors in individual exposure estimates[21,22,23,24] as well as the evaluation of environmental justice[25,26], it is essential to collect mobility-based high-resolution space-time data to understand the association between environmental exposures and health outcomes. A further challenge lies in the spatial mismatch between environmental datasets and individual-level biometric measures collected at the body-worn sensor. While street-view imagery enables fine-grained environmental characterization at the human visual scale, and satellite-derived vegetation indices (e.g., NDVI) offer broader coverage at coarser resolutions (10–250 m), both require explicit spatial alignment strategies that link mobility trajectories to environmental exposures at the exact locations where physiological and behavioural data are recorded. Addressing this scale incongruity requires explicit spatial alignment strategies that link mobility trajectories to environmental exposures at the exact locations where physiological and behavioural data are recorded.

In response to the challenges posed by the MAUP and UGCoP, researchers across multiple disciplines related to environmental health have increasingly taken part in studies on behavioural processes between environmental exposures and health outcomes, by moving toward individual-level mobility-based frameworks[1,27,28,29]. With increased interest in the environment-mobility-health (E-M-H) nexus, existing literature has predominantly concentrated either on the antecedent (environment-mobility) domain[17,30,31,32]or the consequent (mobility-health) domain[33,34,35], while integrative studies encompassing all three domains remain limited. Nevertheless, individuals' daily activities and travel have significantly shaped the dose-response effects of environmental exposures on health outcomes. Where, when and how individuals project, schedule and conduct their activities over 24-h period is closely associated with their exposures to environmental risks and the consequent health outcomes[17,18,36]. Moreover, another critical gap lies in the fact that few studies provide an actionable analytical framework linking real-time monitoring and longitudinal tracking of health outcomes, limiting the ability to generate robust and actionable evidence for treatment and interventions. This being said, addressing these gaps not only improves the accuracy of environment-health assessments but also uncovers the mediating role of daily activities and travel and underlying behavioural processes in the complex E-M-H nexus[37], thereby providing a strong rationale for personalized, mobility-informed treatment and interventions aimed at mitigating exposure risks and promoting health outcomes.

The rise of mobile and wearable technologies[38,39], particularly mobile health (mHealth) devices[40,41,42] and portable environmental sensors[43,44,45], has enabled new approaches to synchronously monitoring individuals' behaviour, environmental exposures and physiological metrics. Through integration with smartphone applications, participants can document their health status, medical histories (e.g., respiratory and cardiovascular diseases), subjective physiological experience and environmental risk perception, enabling more precise and personalized risk evaluation and warning via machine learning-based or rule-based approaches. Here, I introduce the Real-World Multimodal Mobile Monitoring of Behaviour, Respiration, and Exposures (air pollution, noise, light, heat, greenness, etc.) for Treatment and Health Evaluation (M3-BREATHE). The end-to-end protocol provides comprehensive guidance on hardware setup, software configuration, data acquisition procedures, and analytical workflows required for deploying the M3-BREATHE. This protocol is suitable for researchers in environmental health, transport geography and personalized medicine conducting controlled

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clinical experiments (short-term, high-fidelity monitoring) or real-world longitudinal studies (long-term, non-obtrusive monitoring), but not recommended for studies requiring ultra-high-precision laboratory-level respiratory measurement (e.g., invasive pulmonary function testing).

Protocol

The conduct of this protocol for research purposes, either in controlled clinical experiments or in real-world contexts, should be undertaken with prior approval and in accordance with the guidelines of the researchers' institutional human research ethics committee. Informed consent must be obtained from the participant prior to the beginning of the survey. The empirical investigation for this article follows the guidelines of Peking University's Human Research Ethics Committee. All participant-based research using this protocol must obtain prior ethical approval from the investigator's institutional research ethics committee. All participants must provide written informed consent prior to any data collection procedures. Recommended equipment settings are listed in the **Table of Materials.**

1. Wearable mobile health devices setup

1. Chest-worn respiratory sensor setup
    1. Instruct the participant to put on the sensor-integrated shirt or the chest strap directly against their skin, beneath their clothing, to ensure optimal skin contact for electrophysiological signal acquisition.
    2. Adjust the shirt or strap fit so that it is snug and comfortable, ensuring that it does not constrain natural breathing or movement. Proper fit is confirmed when the sensor maintains stable contact with the skin and does not slip or shift noticeably during deep inhalation.
    3. Ensure the sensor is fully charged and successfully paired with the smartphone via Bluetooth to facilitate real-time monitoring and recording of respiratory rate (in breaths per minute, breaths/min), tidal volume (in milliliters, mL), and heart rate (in beats per minute, BPM). Confirm that the device clock is synchronized with the smartphone's system time to enable accurate temporal alignment across multimodal data streams.

        NOTE: Respiratory rate and heart rate can also be derived using wearable wristband devices, particularly for long-term monitoring (see step 1.2). Respiratory rate is indirectly calculated based on respiratory-induced variations in photoplethysmography (PPG) or heart rate variability (HRV) signals (see step 5.2). This alternative approach can help reduce participant discomfort and burden associated with prolonged use of a chest-worn respiratory sensor.
2. Smart wristband device setup
    1. Choose the non-dominant wrist as the primary site for device placement, and secure the wristband firmly on the wrist, proximal to the ulnar styloid process.
    2. Adjust the strap to ensure optimal photoplethysmography (PPG) sensor-skin contact without impairing peripheral circulation. The fit is correct when the strap is snug but allows for the insertion of one fingertip.
    3. Open the companion application on the smartphone and ensure the device is successfully connected via Bluetooth.
    4. Navigate to the real-time display within the app, and confirm that the application settings are configured for continuous monitoring of all desired parameters, including: cardiovascular metrics such as heart rate (HR, units: beats per minute, BPM), heart rate variability (HRV, units: milliseconds, ms), blood pressure (BP, units: millimeters of mercury, mmHg), and electrocardiogram (ECG, units: millivolts, mV); respiratory and metabolic measures such as algorithmically estimated respiratory rate (RR, units: breaths per minute, breaths/min), blood oxygen saturation (SpO_2, units: percentage, %), and metabolic equivalents (METs, ratio); peripheral skin temperature (units: degrees Celsius, °C); motion data from the accelerometer (ACC, units: gravitational acceleration, g).

        NOTE: The PPG sensor on the back of the device must maintain constant and flush contact with the skin. Re-tighten the strap if it gets loose over time.

2. Environmental sensors setup

1. Connect the environmental sensing devices to the portable power bank (10,000 mAh capacity) using appropriate USB cables to secure 24-h operation. Ensure all connections are stable and cables are neatly stowed to prevent accidental disconnection during movement.
2. Place the multi-air-pollutant monitor (measuring particulate matter $PM_{2.5}$ and PM_{10} in μg/m^3, carbon monoxide CO in ppm, nitrogen dioxide NO2, sulfur dioxide SO_2 and volatile organic compounds VOCs in ppb), noise sensor (in decibels, dB) and temperature sensor (in degrees Celsius, °C) in the fishnet side pocket of the participant's backpack. Attach the light exposure tracker (illuminance, in lux, lx; Circadian Light A, CLA, 0-1) firmly to the front-facing side of the backpack's shoulder strap, positioning it to face forward/outward without obstruction.

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3. Instruct the participant to carry the backpack along with their daily activities and travel for 24 h, and to charge the power bank every night. Provide a contact number for any issues.
4. Confirm all sensors are powered on and have successfully established a real-time data stream, logging and transmitting data to the companion smartphone application.

   NOTE: All sensors should be calibrated strictly according to their respective manufacturers' specifications before being handed to the participants. Low-cost air pollution sensors are recommended to undergo a collocation calibration test against a research-grade reference instrument for a period of two weeks at a representative site to derive sensor-specific correction factors. The collocation procedure involves placing the low-cost sensor adjacent to the reference instrument (within 1 m, at the same height), operating both simultaneously for at least 14 days to capture a range of environmental conditions. Paired data are then used to derive linear or non-linear correction models (e.g.,$C_{corrected} = (C_{raw} - b)/a$) via regression analysis. For sensors without access to reference instruments, zero calibration using HEPA-filtered clean air is recommended to correct baseline drift. A 10–15 min stabilization period under clean air conditions is required before recording baseline values. Prior to each deployment, a pre-deployment verification check should be performed to confirm sensor responsiveness and data transmission stability. During data collection, daily quality assurance (QA) logs should be maintained to document any anomalies, sensor drift, or connectivity interruptions. Temperature and relative humidity sensors (RH, units: percentage, %) are usually integrated as components in the air pollution sensing devices. Continuous noise exposure can alternatively be measured using the smartphone's built-in microphone via the dedicated application. Light exposure and heat exposure collected using the smart wristband and mobile phone are not recommended, due to high susceptibility to obstruction and confounding factors such as clothing and placement. For a longitudinal study or practice, an integrated modular device with a small size for multi-pollutant monitoring is recommended.

3. Smartphone application configuration

1. Install the M3-BREATHE mobile application (custom-developed for Android/iOS) on the participant's smartphone.
2. Enable GPS, Bluetooth, and background data synchronization.
3. Configure the app to record GPS location data at 1-min intervals and sync sensor data every 5 min.
4. Set the data sampling rate to 1 Hz (i.e., one sample per second) for each connected device (wearable physiological monitors and portable environmental sensors) to ensure high-temporal-resolution data capture.
5. Confirm that all settings are saved before initiating data recording.

   NOTE: It is good practice to perform a brief (e.g., 5-min) test recording after configuration. Review the test data file to confirm that all expected data streams (GPS coordinates, HRV, ECG, air pollution, noise, etc.) are logged, correctly time-stamped and contain plausible values before commencing the full study period. It is worth noting that greenness exposure is indirectly measured by integrating high-spatiotemporal-resolution GPS trajectory data with geographic datasets such as OpenStreetMap or satellite-derived vegetation indices (e.g., NDVI for vegetation cover). To address the spatial scale mismatch between coarse-resolution environmental layers (e.g., NDVI at 10–250 m) and individual-level biometric measures, the protocol assigns environmental exposure values to each GPS point based on nearest-neighbour or buffer-averaged extraction methods, ensuring spatial alignment at the location and time of each physiological measurement.

4. Data Acquisition Procedures

1. Participant onboarding
   1. Recruit participants using a stratified sampling method to ensure representativeness of the target population.
   2. Provide participants with comprehensive guidance materials, including a detailed instruction sheet, a manual and preferably an instructional video, on the procedures for wearing and using the sensors as well as operating the accompanying smartphone application.
   3. Conduct a baseline health survey within the app to collect demographic data (e.g., gender, age, height, weight, marital status), medical history (e.g., non-communicable diseases, NCDs), respiratory health status (e.g., smoking pack-years, chronic cough or dyspnea), and other self-reported health information (e.g., general physical and mental health, alcohol consumption, dietary patterns, physical activity levels, sleep quality).

      NOTE: Recommending standardized instruments for self-reported health information collection, including the 12-Item Short Form Survey (SF-12) for general health[46], the Alcohol Use Disorders Identification Test-Consumption (AUDIT-C)[47], the International Physical Activity Questionnaire-Short Form (IPAQ-SF)[48], and the Pittsburgh Sleep Quality Index (PSQI)[49]. Alternatively, 7-item Likert scales can be applied for convenience purposes.

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2. Daily Monitoring Routine
    1. Instruct the participant to wear the mobile-health physiological sensors and carry the portable environmental monitoring devices (backpack with a fishnet pocket) for a continuous 24-h period. During sleep, the environmental sensors should be positioned in the bedroom to maintain exposure assessment, with particular attention to the light sensor.
    2. Remind the participant to recharge the power bank each evening and keep it connected to the environmental sensors to maintain uninterrupted operation and data.
    3. The data stream will be monitored either through manual checks by the research team every 4 h, or preferably, by an automated alert from the backend system within the application, triggered after 30 min of data sync interruption. In case of disconnection for over 30 min, the participant will be contacted to address the issue.
    4. Encourage the participant to complete an activity-travel diary survey within the application (a template is provided in **Table 1**) each day in the evening, to enhance the accuracy of daily activity and mobility contextualization for the sake of more effective and targeted treatment and intervention. An automated reminder (application/SMS) is sent each evening at 8 p.m. to promote consistent compliance.

        NOTE: An interactive interface, featuring a map for trajectory (with timestamps) visualization, can be deployed with machine learning-based identification of activities and trips based on GPS data. Ideally, advice or reminders can be sent automatically through the application to the participant to record an activity or a trip, proximate to the occurrence of the next activity/trip episode, to mitigate end-of-day recall bias. A final review before bedtime is recommended to consolidate the daily record. In addition, ecological momentary assessment (EMA) can be utilised to capture subjective experiences (e.g., breathlessness, fatigue, environmental pollution perceptions) three times per day (e.g., 8 a.m., 12 p.m., and 4 p.m.) via push notifications.
3. Data Synchronization and Backup
    1. All collected real-time and activity-travel diary data are automatically synced to a secure, encrypted cloud server via Wi-Fi or cellular network. The cloud database is configured with regular automated backups to prevent data loss.
    2. All data is first stored on the smartphone's internal memory. Synced data is automatically cleared from the device every day after successful verification by the cloud server.
    3. All data are anonymized at the point of collection and safely stored on institutional servers. The storage, transmission and disposal of the data are managed in strict accordance with the data security and privacy policies of the host institution.

        The summary of data and the source are provided in **Table 2.**

ParticipantID	Date	StartTime	EndTime	Activity(A)/Trip(T)	Activity Type/ Trip Mode
1	13/08/2025	0:00	7:00	A	Sleep
1	13/08/2025	7:00	7:30	A	Breakfast
1	13/08/2025	7:30	8:00	T	public transport
1	13/08/2025	8:00	11:30	A	Work
1	13/08/2025	11:30	11:45	T	Walk
1	13/08/2025	11:45	12:45	A	Lunch
1	13/08/2025	12:45	13:00	T	Walk
1	13/08/2025	13:00	17:00	A	Work
1	13/08/2025	17:00	17:15	T	Bicycle

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1	13/08/2025	17:15	18:00	A	Cooking
1	13/08/2025	18:00	19:00	A	Supper
1	13/08/2025	19:00	21:00	A	Leisure
1	13/08/2025	21:00	21:30	A	Bath
1	13/08/2025	21:30	23:59	A	Sleep
1	14/08/2025	0:00	6:30	A	Sleep
1	14/08/2025	…	…	…	…

Table 1: Activity-travel diary template for M3-BREATHE monitoring. The table illustrates how behavioral data are recorded and organized for integration with exposure and health analyses.

Wearable mobile health devices	Smart phone application	Portable environmental sensors
Respiratory Rate (RR)	GPS coordinates	Particulate Matter (PM2.5, PM10)
Tidal Volume (TV)	Activity-travel diary	Carbon Monoxide (CO)
Heart Rate (HR)	Ecological momentary assessment (EMA)	Nitrogen Dioxide (NO2)
Heart Rate Variability (HRV)	Baseline health information	Sulfur Dioxide (SO_2)
Electrocardiogram (ECG)	Age, gender, etc.	Volatile Organic Compounds (VOCs)
Blood Pressure (BP)	Body Mass Index (BMI)	Noise (N)
Blood Oxygen Saturation (SpO_2)	Socioeconomic status (SES)	Temperature (T)
Metabolic Equivalents (METs)	Medical history	Relative Humidity (RH)
Skin Temperature (ST)	Respiratory health status	Illuminance (lx)
Accelerometer (ACC)	Self-reported health information	Circadian Light A (CLA)

Table 2: Summary of data and the source of devices. The table lists key metrics (e.g., heart rate, respiratory rate, particulate matter, temperature) and their corresponding measurement modalities.

5. Data Processing and Analysis

1. Device calibration and quality assurance

   Prior to deployment, all devices will undergo basic calibration checks according to manufacturer specifications. For environmental sensors (e.g., $PM_{2.5}$ monitors), this may include zero calibration or comparison against reference instruments when available. During data processing, quality assurance and quality control (QA/QC) procedures will be applied to both environmental and biometric data streams. All QA/QC steps will be documented to ensure transparency and reproducibility.

   1. Range checks to remove physiologically or physically implausible values (e.g., extreme heart rate or pollutant concentrations).
   2. Detection and exclusion of signal artifacts (e.g., motion-related noise in wearable data). Smoothing or filtering procedures (e.g., moving averages) where appropriate.

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3. Identification and handling of missing or sparse data segments.

2. Preprocessing
    1. Raw sensor data are cleaned (GPS drift is corrected through the Kalman filter) and synchronised with activity-travel diaries using the Multimodal Temporal Alignment approach via Python-based preprocessing packages (e.g., Pandas, GeoPandas, Tsfresh, Pyo3). All processed data are subsequently integrated into the M3-BREATHE database.
    2. To ensure temporal alignment between environmental exposures and biometric measurements, synchronize all data streams to a unified timestamp framework. Standardize device clocks prior to deployment using network-based time protocols. During data processing, align timestamps from different devices to a common reference time (e.g., UTC), and correct clock drift using linear interpolation or time-offset adjustment.
    3. Aggregate higher-frequency data (e.g., wearable-derived heart rate) to match the resolution of environmental measures or adjust environmental data to match higher-frequency data, depending on the analysis objective. Handle missing timestamps or irregular sampling intervals by applying interpolation or excluding data based on predefined quality criteria.
    4. Process GPS data alongside external geographical datasets characterizing the built and natural environments to contextualize individual exposures. Compute greenness exposure by integrating GPS records with OpenStreetMap data and satellite-derived vegetation indices (e.g., NDVI). Address scale disparities between high-frequency GPS trajectories and lower-resolution satellite-based environmental layers by applying nearest-neighbour or buffer-based spatial alignment methods.
    5. For each GPS fix, assign the corresponding environmental exposure value from the intersecting or nearest pixel, or calculate a weighted average of pixels within a defined buffer radius (e.g., 10 m, 30 m) that reflects typical GPS spatial uncertainty. Ensure that environmental exposures are attributed at a spatiotemporal resolution compatible with the simultaneously collected physiological and behavioural data.
3. Feature Extraction
    1. Summarize environmental and mobile health monitoring data using key metrics, including measures of central tendency (mean, median, weighted mean), extreme values (maximum, minimum, percentiles), variability (standard deviation, variance, coefficient of variation, interquartile range), and temporal patterns (peak frequency, duration above threshold, rate of change, diurnal, weekly, or seasonal variation).
    2. Calculate cumulative air pollutant exposure using total exposure metrics (average concentration and inhaled dose) and the area under the curve (AUC). Compute the inhaled pollutant dose (D) over an exposure duration (t) using the relation below.

$$D = C \times \dot{V}_E \times t$$

with minute ventilation defined as:

$$\dot{V}_E = RR \times V_T$$

where C is the pollutant concentration in the inhaled air (μg/m^3, ppb or ppm, depending on unit convention); $\dot{V}_E$ is the minute ventilation, i.e., the total volume of air inhaled per minute (L/min); RR is the respiratory rate (breaths/min); V_T is the tidal volume, the volume of air inhaled per breath (L/breath); and t is the exposure duration (min).

Note that the dose D is calculated in μg, provided that $\dot{V}_E$ is converted to m^3/min (i.e., divide by 1000) to maintain unit consistency.

    3. Given that chest-worn respiratory sensors are usually expensive and burdensome to wear continuously for extended periods in real-world contexts, compared with monitoring durations of several hours to a few days in controlled clinical experiments, assess respiratory rate using accelerometer and gyroscope data obtained from the smart wristband by applying machine learning algorithms[50,51]. To improve the accuracy of the assessment, a random forest model usually acts as a filter, rejecting data that result in an inaccurate respiratory rate reading, after which a Convolutional Neural Network (CNN) model can be used to extract the respiratory rate[52,53].
    4. Extract behavioural features from GPS data, smartphone-based travel-activity diaries, and motion data from wristband accelerometers. Generate a set of individual behavioural indicators to analyze associations among environmental exposure, behaviour, and health outcomes. **Table 3** summarizes multi-scale profiles across the behavioural, environmental exposure, and health domains.

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NOTE: Arrows illustrate the sequential alignment of profiles across behaviour, environmental exposure and health domains.

4. Data Analysis
    1. Descriptive and exploratory analyses
        1. Conduct initial descriptive statistics to characterize the distribution, central tendency, and variability of all variables.
        2. Perform correlation analysis and variance inflation factor (VIF) tests.
        3. Visualize exposure and health data using time-series, boxplots, and geospatial heatmaps.
    2. Acute associations and temporal lags
        1. Use distributed lag models (DLMs).
        2. Bin exposure data into time blocks.
        3. Conduct sensitivity analyses.
        4. Apply time-series methods (ARIMA, VAR).
    3. Chronic associations and cumulative exposures
        1. Aggregate exposure metrics.
        2. Link exposure to repeated health assessments.
        3. Use mixed-effects regression models.
        4. Apply longitudinal or survival analysis.

NOTE: It is necessary to consider time-window strategies. **Table 4** compares different time-window options.

    4. Behavioural mediation and moderation analyses
        1. Explore mediation using SEM or causal analysis.
        2. Examine moderation effects.
    5. Exposure-response functions and dose-response curves
        1. Quantify exposure-response relationships.
        2. Estimate inhaled dose-response functions.
        3. Investigate non-linear associations.
    6. Longitudinal and life-course analyses
        1. Evaluate chronic effects.
        2. Explore life-course trajectories.
    7. Personalized treatment and intervention
        1. Construct personalized profiles.
        2. Identify high-risk episodes.
        3. Recommend medical evaluations.
    8. Population health intelligence
        1. Generate exposure maps.
        2. Analyze behavioural trends.
        3. Assess health inequalities.
        4. Use machine learning models.

Behavioural Domain	Environmental Exposure Domain	Health Domain
Transient/momentary features	**Instantaneous exposure profiles**	**Short-term health profiles**
Accelerometer vector magnitude	Real-time exposure	Cardiorespiratory stress responses
Instantaneous activity intensity	Acute exposure	Acute changes in HR, BP, HRV, SpO_2,etc.
Episode-level/bout-level features	**Episode-level exposure profiles**	ECG arrhythmic events
Start time, end time, duration	Average pollutant concentration	Respiratory symptoms and function
Activity type	Cumulative pollutant exposure dose	Episode-specific ventilation rate fluctuations
Location type	Peak exposure	Self-reported coughing and shortness of breath
Indoor/outdoor	Percentile-based exposure	Psychological and wellbeing states

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Transport mode	Exposure variability	EMA mood, stress, fatigue, etc.
Trip distance	Above-threshold exposure duration	Episode-level cognitive and affective strain
Trip average speed	AUC exposure	Acute exposure burdens
Trip purpose	Indoor vs. outdoor exposure	**Medium-term health profiles**
Physical activity intensity	Transport microenvironment exposure	Subclinical cardiovascular markers
Respiratory pattern	Workplace/home/recreational exposure	Sustained increases in resting HR or BP
Inter-segment and daily-level features	**Daily-level exposure profiles**	Early hypertension risk scores
Sedentary breaks and transitions between activity states	Daily mean exposure	Respiratory health trends
Transitions between indoor and outdoor environments	Daily cumulative exposure dose	Frequency of respiratory symptom episodes
Transitions between transport microenvironments	Daily peak exposure	Decline in lung function (RR, SpO_2, etc.)
Activity–trip chains	Frequency/duration of daily peaks	Metabolic and physical activity status
Activity/tour complexity	Daily minimum exposure	Cumulative MET-min/week
Activity-travel Chain	Daily percentile exposure	Activity intensity distribution
Time-use structure	Daily above-threshold exposure duration	Sleep and recovery health
Activity space, radius of gyration, spatial entropy	Daily AUC exposure	Fatigue persistence beyond daily variation
Overall travel direction	Daily exposure variability	Mental health and wellbeing
Time in Moderate-to-Vigorous Physical Activity (MVPA)	Daily exposure slope (rate of change)	Stress reactivity, anxiety scores, etc.
Circadian rhythm	Daily peak-to-mean ratio	Cumulative exposure risks
Habitual routine patterns	**Longitudinal exposure profiles**	Average pollutant concentration and dose
Routine daily activity-travel patterns	Long-term mean exposure	**Long-term health profiles**
Variability in daily activity-travel patterns	Long-term cumulative exposure dose	Cardiovascular disease risk and progression
Chronotype of circadian rhythm	Peak exposure	Respiratory and pulmonary outcomes
Weekend/weekday discrepancies in circadian rhythm	Frequency/duration of peak events	Metabolic and obesity-related outcomes

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Regularity of Mobility (trajectory, commute, mode, etc.)	Percentile-based exposure	Mental health outcomes
Long-term trends in physical activity	Time above threshold	Systemic health and biological aging
Sedentary and active behavior cycles	AUC exposure	Functional and quality-of-life outcomes
Dietary timing and regularity	Daily/weekly/monthly/seasonal variability	Life-course health trajectories
Adaptive behavior under environmental extremes	Exposure time-series mining	Population health intelligence

Table 3: Multiscale data profiles of M3-BREATHE. The table highlights how behavioural metrics relate to short-, medium-, and long-term health outcomes.

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Window	Use-Case	Modeling Approach	Pros	Cons
1 hour	Very acute effects	Include 1-h moving average as covariate in LME or time-series; or use DLNM with hourly lags	Captures immediate peaks; minimal dilution	High noise; needs high-res data; limited chronic insight
24 hours	Daily (short-term)	24-h moving avg in mixed model; DLNM or DLM with 1-day lags	Reduces noise; aligns with regulatory metrics	May smooth out peak events
7 days	Weekly (sub-acute)	7-day avg covariate; DLM with lags up to 7 days	Captures accumulated short-term exposure	Blurs daily variability; moderate smoothing
30 days	Monthly (medium-term)	30-day avg; DLNM with monthly lag structure	Reflects recent exposure load; smoother	Misses day-to-day spikes; assumes linear cumulation
90 days	Seasonal (long-term)	90-day avg or distributed lag across ~90d	Captures seasonal trend	Over-smoothes; confounds with seasonal factors
365 days	Annual (long-term)	1-year avg in LME or Cox model	Useful for chronic risk; controls for season	Ignores acute variation; confounded by aging trends

Table 4: Time-window strategies and analytical approaches. Summary of time-window strategies and corresponding analytical approaches used to assess environmental exposures across different temporal scales, highlighting their applicability, strengths, and limitations.

Results

The following representative results are drawn from a pilot study that integrated portable air pollution sensors, activity–travel diaries and episode-level self-reported subjective wellbeing (SWB). Participants were required to carry portable air pollution sensors during their daily activities and travel for a continuous 48-h period (either Friday–Saturday or Sunday–Monday) and to complete accompanying activity-travel diaries. Data from a total of 112 valid participants, from December 2017 to February 2018, in the Meiheyuan residential community, Beijing, were included in the analysis. This preliminary analysis focuses on the interrelations among $PM_{2.5}$ exposure (environmental domain), daily activities (behavioural domain), and cognitive SWB (health domain). While this pilot does not yet validate the full M3-BREATHE pipeline, particularly the integration of physiological signals (e.g., HRV, respiratory rate) and real-time exposure alerts, it serves as an illustrative example of the analytical approach. Full end-to-end validation of the complete multimodal system is needed for future work.

To validate the end-to-end M3-BREATHE pipeline, we assessed data integrity (99.3% GPS data capture rate and 93.2% air pollution monitor capture rate), synchronization accuracy (≤1 s time asynchrony between multimodal data), and calibration validity ($PM_{2.5}$ measurements from low-cost sensors had a Pearson correlation of r=0.86 with research-grade reference instruments after collocation calibration). The pipeline integrated

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GPS trajectory data, environmental exposure data, and self-reported diary data, enabling the identification of activity-specific exposure patterns and exposure-health associations. These results confirm the reproducibility and reliability of the M3-BREATHE protocol for real-world E-M-H monitoring.

Figure 1 represents the average $PM_{2.5}$ exposure concentration (blue bars, left axis) and cognitive SWB scores (orange line, right axis) across a range of daily activities and travel. The results show that $PM_{2.5}$ exposure concentrations differed markedly by activity types. The highest average exposure was observed during sleep, indicating high-level air pollution concentration indoors at home. This may be because the study area is a comparatively highly polluted area compared with other parts of Beijing. Other activities with consistently high exposure concentrations included surfing the Internet, entertainment, studying, and taking a walk, all of which exceeded 20 μg/m^3. Mid-range exposure concentrations were found for activities such as work, physical activity, caregiving and trips (15–20 μg/m^3). The lowest exposures were recorded during personal affairs, seeing a doctor, dining and personal care (5–15 μg/m^3). These differences, based on this pilot study, reflect the specific environmental contexts in which activities take place (indoor vs. outdoor and sedentary vs. active). In contrast, cognitive SWB scores showed less variability across activities, ranging narrowly between 1.7 and 2.3 (on a 1–5 dissatisfaction scale). The lowest satisfaction scores were associated with seeing a doctor and trips, followed by personal affairs and work, while the highest satisfaction scores were linked to physical activity, entertainment and social activity, which are often intrinsically rewarding activities.

It is found that the activity types associated with higher $PM_{2.5}$ exposure concentrations did not always correspond to lower cognitive SWB, suggesting decoupling patterns between $PM_{2.5}$ exposure and self-reported cognitive SWB. For instance, physical activity and entertainment combined with elevated exposure and high cognitive SWB scores. This may reflect that the health risk of $PM_{2.5}$ exposure may be overlooked or not consciously linked to cognitive SWB, probably due to the invisible nature of $PM_{2.5}$, which makes it even more insidious and difficult to detect in the environment. By comparison, pollutants like SO_2, NO_x and O_3 have more readily detectable characteristics. For instance, SO_2 has a distinctive sulfuric odor and NO_x is often associated with visible emissions from industrial processes and vehicle exhaust. These gaseous pollutants can cause immediate irritation to the eyes, throat, and respiratory system, making them easier to detect and respond to without the need for specialized equipment. However, $PM_{2.5}$ is the most dangerous pollutant because it can penetrate into the gas exchange regions of the lung (alveolus), and very small particles may pass through the lungs to affect other organs and hence do harm to the circulatory system[59]. This finding highlights the unique challenge posed by $PM_{2.5,}$ whose imperceptible nature and severe health impacts make it a critical but often underestimated environmental risk across various daily activities.

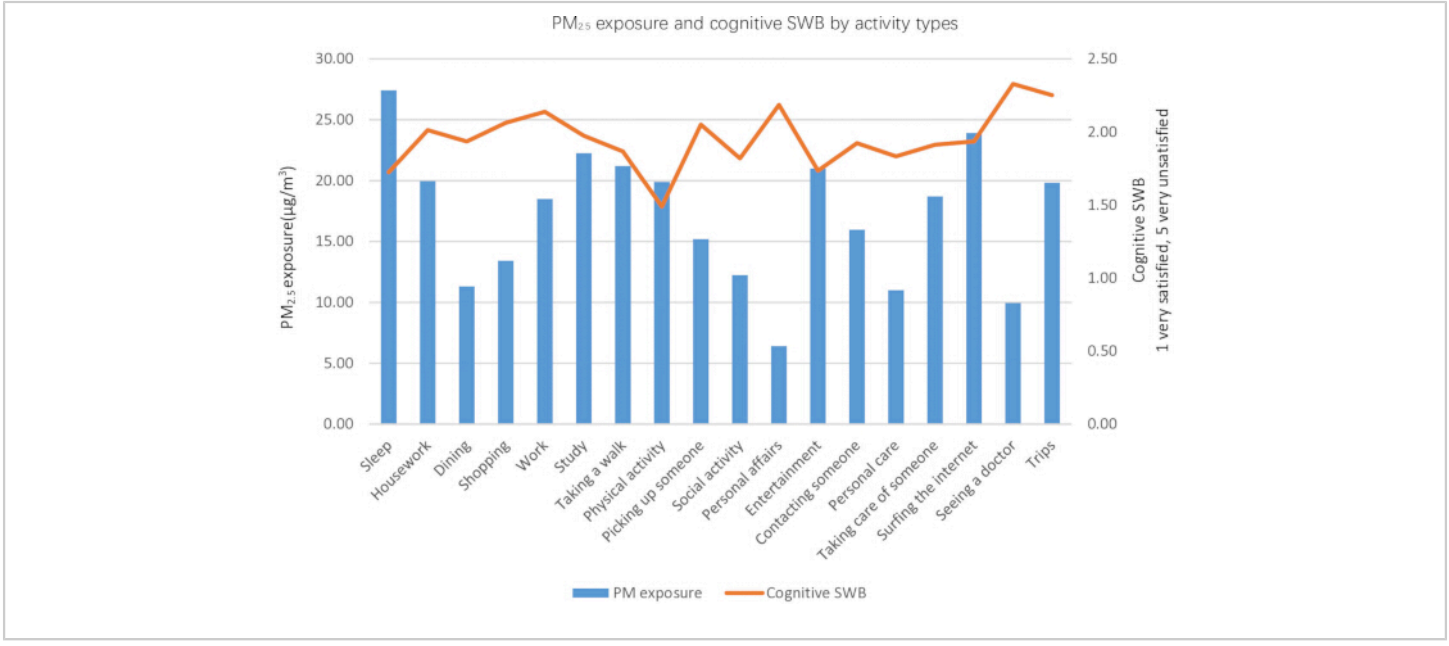


Figure 1: $PM_{2.5}$ exposure concentration and cognitive subjective well-being (SWB) by daily activity type. Cognitive subjective well-being (SWB) is measured on a Likert scale ranging from 1 (very satisfied) to 5 (very unsatisfied). The figure illustrates variations

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in fine particulate matter ($PM_{2.5}$, $\mu g/m^3$) exposure levels and corresponding well-being scores across activities. Data points represent observed values. Please click here to view a larger version of this figure.

To examine the relationship between air pollution exposure and subjective well-being (SWB), this pilot study employs a Generalized Additive Model (GAM), allowing for flexible nonlinear effects. Both contemporaneous air pollution exposure (AAP) and lagged exposure from the previous activity (AAP_1) are modeled using smooth functions, while socio-demographic characteristics (gender, age, hukou, education, marital status, and income) are included as control variables. The results show that neither AAP nor AAP_1 has a statistically significant effect on SWB, with p-values of 0.234 and 0.266, respectively. Although the estimated smooth terms suggest some nonlinear patterns, these effects are not robust (**Figure 2**). In contrast, certain individual characteristics exhibit stronger associations with SWB.

These findings suggest that short-term air pollution exposure, whether contemporaneous or lagged by one activity episode, does not significantly influence SWB. One possible explanation is that individuals may not immediately respond to short-term fluctuations in air quality, or that such effects require longer or cumulative exposure to become detectable. Future research could extend this analysis by incorporating multiple lag structures (e.g., exposures from several preceding activity episodes), using time-based measures (e.g., minute-level exposure duration), or applying alternative modeling approaches such as distributed lag models or machine learning methods to better capture dynamic and cumulative effects of air pollution on well-being.

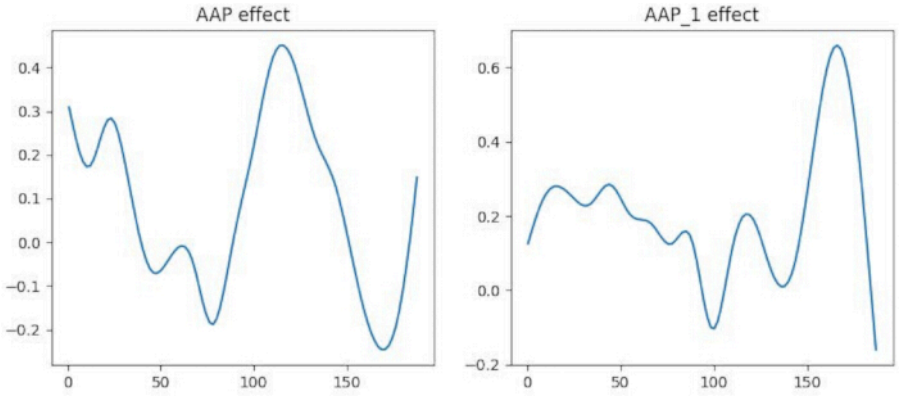


Figure 2: Nonlinear effects of current and lagged air pollution exposure on subjective well-being. The curves represent model-derived effect estimates over time, highlighting short-term fluctuations and delayed responses. AAP denotes contemporaneous exposure, while AAP_1 represents exposure lagged by one time unit (e.g., 1 day). Please click here to view a larger version of this figure.

Discussion

A critical step in deploying the M3-BREATHE protocol is the reliable setup and synchronisation of multimodal devices, as data integrity depends heavily on stable connections between wearable sensors, portable environmental monitors and the smartphone application. It is also essential to ensure all the sensors are calibrated in accordance with the manufacturers' specifications before initiating the survey. Ideally, the portable environmental devices, particularly the air pollution monitor, should be collocated and calibrated against research-grade reference instruments prior to deployment, to ensure the validity and accuracy of the measurements for research purposes.

In controlled clinical experiments or short-term, small-scale surveys, the use of a chest-worn respiratory strap provides high-fidelity respiratory parameters (respiratory rate, tidal volume, ECG), offering greater accuracy than wrist-based sensors and serving as a triangulation basis for validating their algorithmic estimates. In contrast, for longitudinal monitoring in naturalistic environments, including voluntary consumer use, the combination of a smart wristband and smartphone application is preferable, as these devices are less obtrusive and promote sustained participant compliance. For studies with limited resources or participant burden concerns, a minimum kit consisting of a smart wristband, smartphone application, and a single pollutant monitor is recommended. Additional sensors (e.g., chest-worn respiratory strap, dedicated light tracker) should be considered based on specific research objectives and budget. Detailed recommendations for the adaptation of the M3-BREATHE protocol in controlled clinical experiments and real-world longitudinal study/practice are provided in **Table 5**.

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Data Domain	Metric (Abbreviation)	**Controlled clinical experiment**				**Real-world longitudinal study/practice**		
		Chest-worn Sensor	Smart Wristband	Portable Environmental Monitor	Smartphone app	Smart Wristband	Portable Environmental Monitor	Smartphone app
Physiological	Heart Rate (HR)	OOO	OO			OOO		
	Heart Rate Variability (HRV)	OOO	OO			OOO		
	Respiratory Rate (RR)	OOO	O			OOO		
	Tidal Volume (TV)	OO	O			OO		
	Electrocardiogram (ECG)	OOO	OO			OOO		
	Blood Pressure (BP)		OOO			OO		
	Blood Oxygen Saturation (SpO_2)		OOO			OO		
	Metabolic Equivalents (METs)		OOO			OO		
	Skin Temperature (ST)		OO			O		
	Accelerometer (ACC)		OO			OOO		

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Environmental	Particulate Matter (PM2.5, PM10)			OOO			OOO	
	Carbon Monoxide (CO)			OOO			OO	
	Nitrogen Dioxide (NO2)			OOO			OO	
	Sulfur Dioxide (SO_2)			OO			O	
	Volatile Organic Compounds (VOCs)			OO			OO	
	Noise (N)			OOO	O		O	OO
	Temperature (T)			OOO			OO	
	Relative Humidity (RH)			OO			O	
	Light Exposure (illuminance)			OOO			OO	
	Light Exposure (Circadian Light A)			OO			O	
	Greenness				OOO			O

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Behavioral & Contextual	GPS Location / Trajectory				OOO			OOO
	Activity-Travel Diary (ATD)				OOO			O
	Ecological Momentary Assessment (EMA)				OO			O
	Baseline Characteristics (BC)				OOO			OOO
Notes: OOO, minimum kit; OO, standard kit; O, alternative								

Table 5: Recommendations for the adaptation of the M3-BREATHE protocol. Summary of measurement domains and instrumentation levels across study settings (OOO minimum; OO standard; O alternative).

Several modifications and troubleshooting strategies are required when applying this protocol in different settings. First, in case of a disconnection between the monitoring devices and the smartphone or periods over 30 min with null or invalid data (e.g., health parameter data loss due to sensor malfunction or lack of skin contact), automated reports to investigators should be generated so that participants can be contacted promptly to resolve the issue. Investigators can also conduct regular data integrity checks. Second, for longitudinal deployments of the protocol, device adherence and battery management may present greater challenges. Participants may occasionally forget to wear the smart wristband, recharge power banks or fill in the activity-travel diaries. Automated in-app reminders, backend monitoring, and daily diary prompts are effective mitigation tools.

Despite its advantages, M3-BREATHE has several limitations. While smart chest straps yield precise respiratory measurements, they are generally impractical for long-term real-world monitoring due to participant burde[60]. Wristband-based measurements and algorithms for respiration assessment are therefore recommended, though they have reduced accuracy for estimating tidal volume or detecting subtle arrhythmias, and their performance can be affected by motion artefacts[61]. In addition, the utilisation of multiple environmental monitoring devices increases the possibility of potential connectivity failures, non-synchronous clocks or sensor malfunctions. An integrated modular device for multi-pollutant monitoring or alternative approaches (e.g., smartphone-based noise monitoring) should be developed, with the objective of maintaining measurement accuracy while minimizing device size and maximizing portability[62]. Moreover, reliance on in-app self-reported systems (e.g., activity-travel diaries, medical records) may introduce recall bias and increase long-term compliance burdens[63]. Machine learning-based identification of activities and trips can be adopted[64], while tracking medical treatment records from healthcare providers can be seamlessly connected to the smartphone application via a simple one-click interface.

This visionary real-world M3-BREATHE establishes a scalable, non-invasive, and cost-efficient platform for longitudinal environmental-mobility-health (E-M-H) nexus research and practice, advancing existing approaches. Traditional spatial epidemiology often relies on fixed-site monitoring or residential proxies for exposure assessment, making it vulnerable to the modifiable areal unit problem (MAUP) and the uncertain geographic context problem (UGCoP)[21,65]. By integrating multimodal high space-time resolution behavioural, physiological and environmental exposure data, this system moves toward frame-independent individualised exposure assessments, facilitating the unpacking of the behavioural "black box" in the environmental–health relationship[66,67]. It supports both individual and population-based in-depth interpretation of dose–

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response dynamics, linking real-time multi-pollutant exposures to physiological reactivity and health outcomes, enabling personalized treatment and intervention, public health surveillance and translational research linking environmental exposure, daily activities and mobility and health outcomes[68], as well as facilitating individuals' health self-management and risk mitigation for environmental exposures[69] (e.g., daily activity planner in avoiding high-risk exposure based personal historical E-M-H profiles).

Future applications of M3-BREATHE include (1) integration with AI-driven real-time intervention systems, where personalized exposure alerts trigger adaptive behaviour recommendations (e.g., commuting rerouting during peak $PM_{2.5}$ exposure); (2) deployment in vulnerable populations (e.g., older adults, asthmatics) to assess disease-specific exposure-health associations and support precision medicine; (3) combination with genomic data to uncover gene-environment interactions in chronic disease development; (4) miniaturization of integrated modular environmental sensors to reduce participant burden and improve long-term adherence; (5) cross-city and cross-country scaling to generate global E-M-H datasets for comparative environmental health research. In addition, the protocol can be adapted for monitoring other environmental risks by replacing relevant portable sensors.

Disclosures

The author declares no competing interests.

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